Equations of circles

1 **Graph B**

2

a No solution provided

b Quadrilateral

c Circle

3

a

i $(0, 0)$

ii 2 units

b

i $(3, 2)$

ii 3 units

c

i $(1, -2)$

ii 8 units

d

i $(-3, -1)$

ii 5 units

4 Every point on the circle is exactly 5 units away from the point $(0, 0)$.

5

a 1

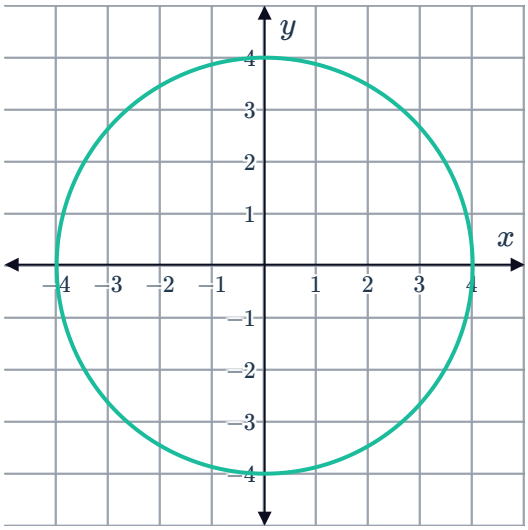
b 0

6 $x^2 + y^2 = 49$

7

a

i

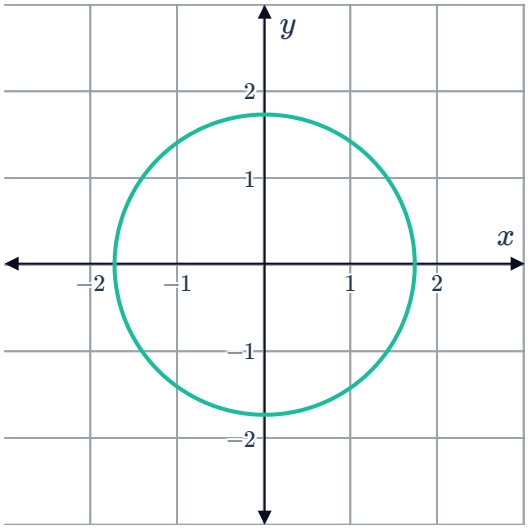


ii $x^2 + y^2 = 16$

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b

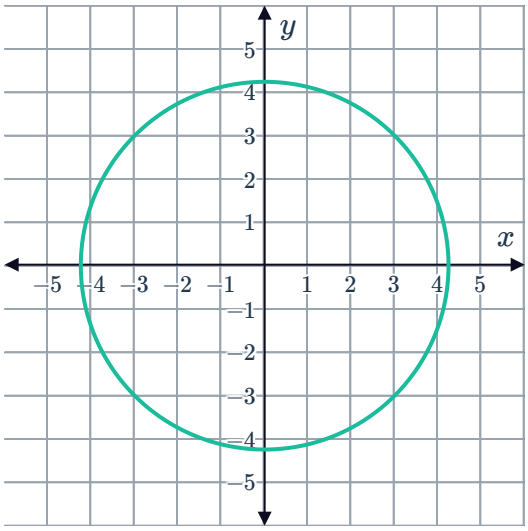
i



ii $x^2 + y^2 = 3$

c

i



ii $x^2 + y^2 = 18$

8

a $(0, 0)$

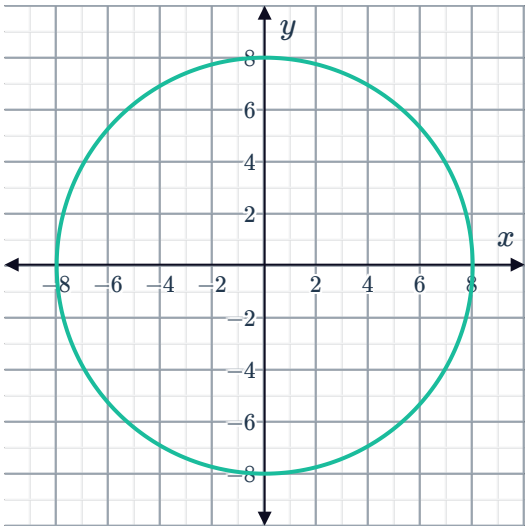
b 8 units

c $x = 8, -8$

d $y = 8, -8$

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e



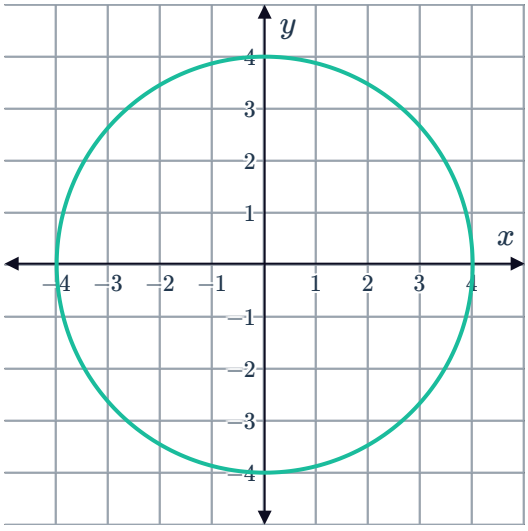
9 $4\sqrt{2}$ units

10

a

i 4 units

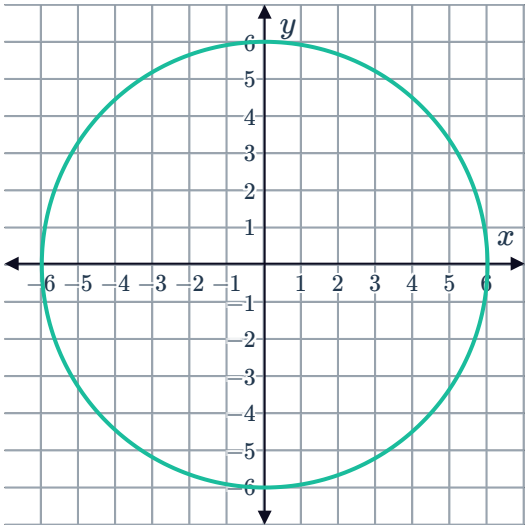
ii



b

i 6 units

ii



11 $x^2 + y^2 = 4$

12

a $5\sqrt{5}$ units

b $x^2 + y^2 = 125$

13

a $x^2 + y^2 = 36$

b 36π units^2

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14 **a** 14 units **b** $y = \pm\sqrt{48}$

15 Inside

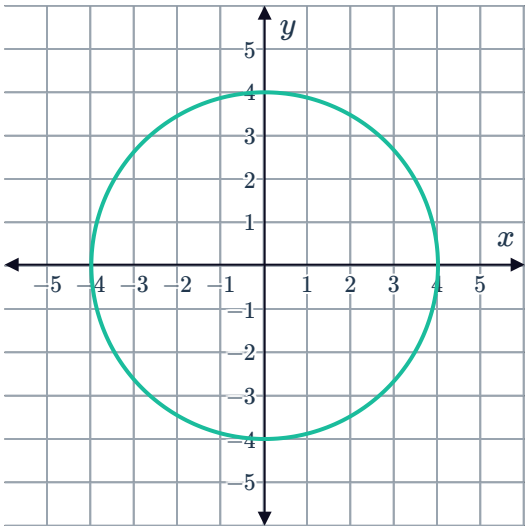
16

a

i $(0, 0)$

ii 4 units

iii

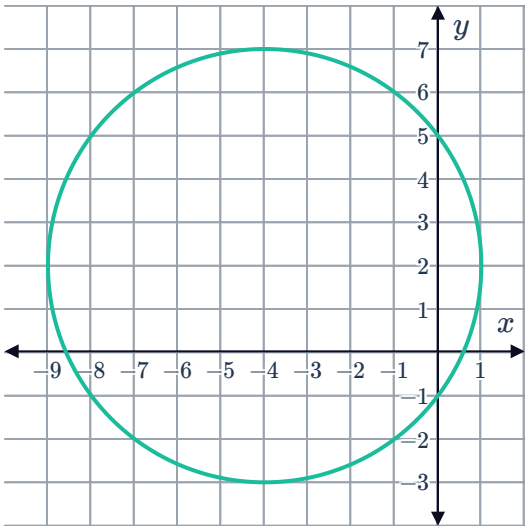


b

i $(-4, 2)$

ii 5 units

iii

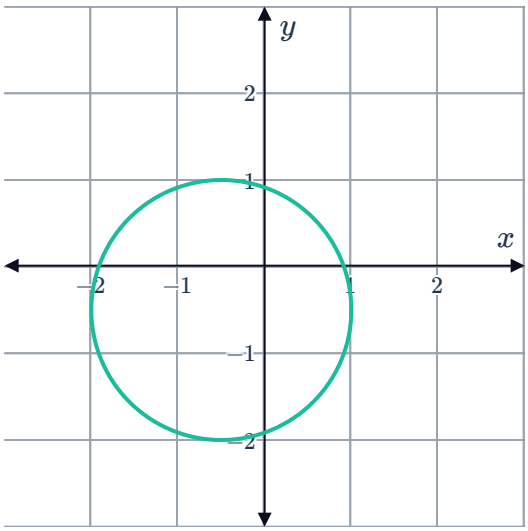


c

i $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

ii $\frac{3}{2}$ units

iii

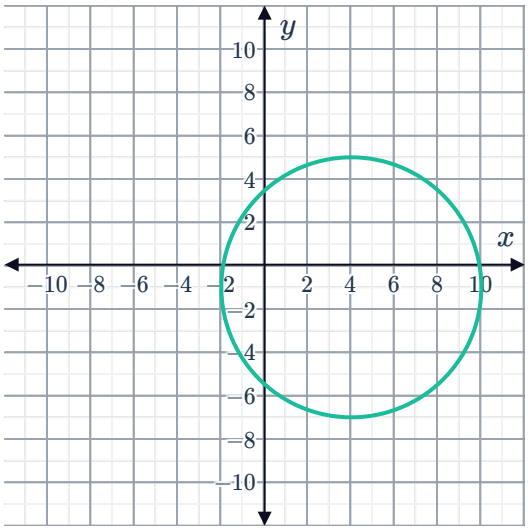


d

i $(4, -1)$

ii 6 units

iii



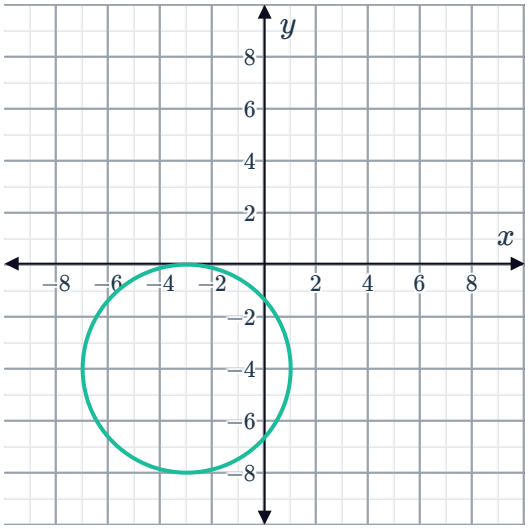
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e

i $(-3, -4)$

ii 4 units

iii

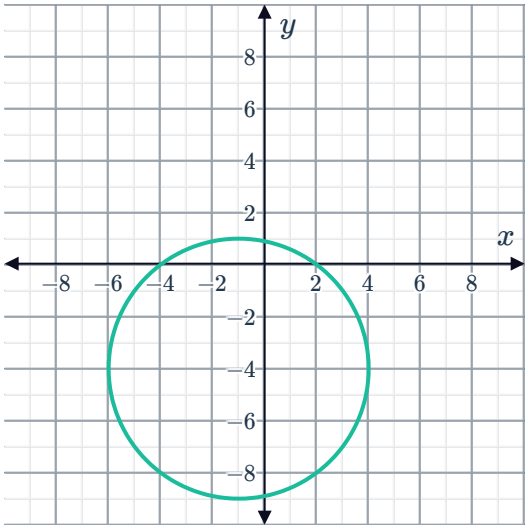


f

i $(-1, -4)$

ii 5 units

iii

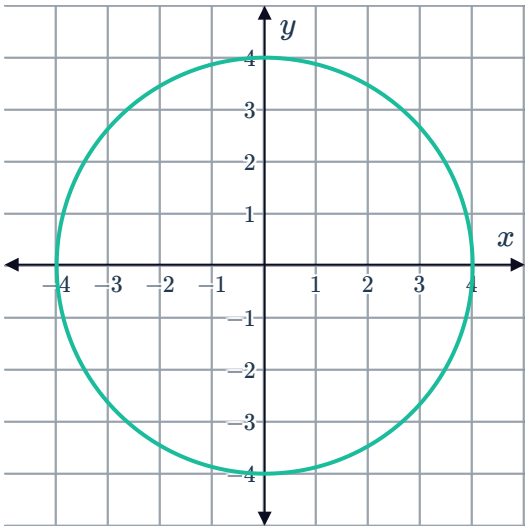


g

i $(0, 0)$

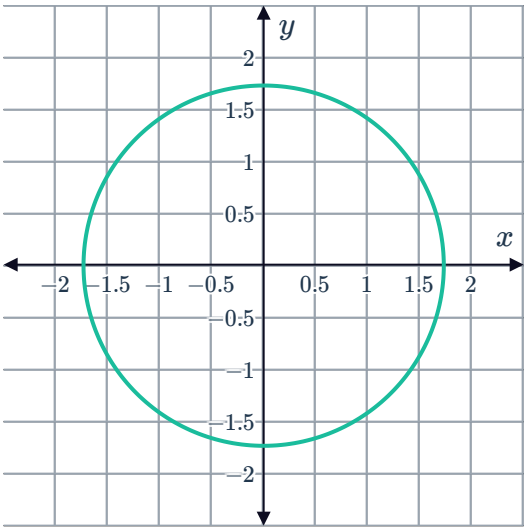
ii 4 units

iii



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17



18 $\sqrt{353}$ units

19

a

i $\left(-3.1, \frac{3}{2}\right)$ ii $\frac{3}{4}$ units

b

i $(6, 6)$ ii $2\sqrt{3}$ units

c

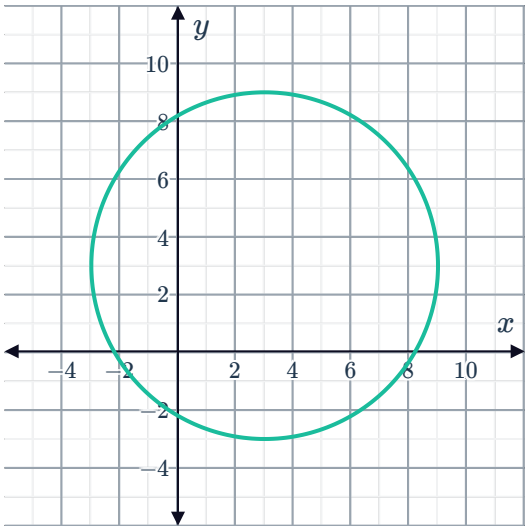
i $(0.7, -4.4)$ ii $\sqrt{3}$ units

d

i $(8, -1)$ ii $\sqrt{13}$ units

20

a



b $(x-3)^2 + (y-3)^2 = 36$

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21

a

i $(-3, -3)$

ii 3 units

iii $(x+3)^2 + (y+3)^2 = 9$

b

i $(1, 3)$

ii 6 units

iii $(x-1)^2 + (y-3)^2 = 36$

22

a $(x-1)^2 + (y+3)^2 = 25$

b $(x+5.7)^2 + y^2 = 9$

23 Two

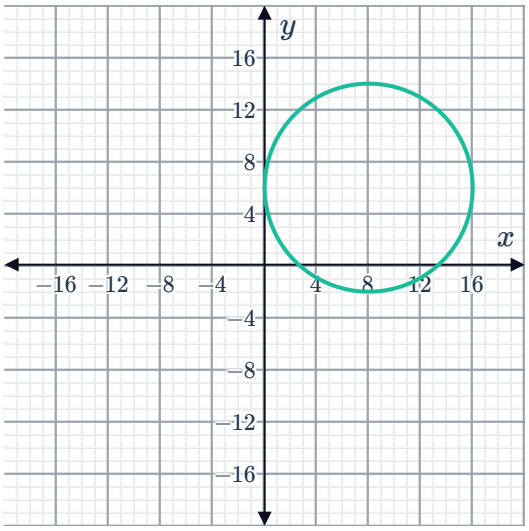
24

a $(x-8)^2 + (y-6)^2 = 64$

b No

25

a



b $x^2 - 4x + y^2 + 12y + 31 = 0$

26

a Outside **b** On **c** Inside

27

a Inside **b** On **c** On

Additional questions

28 $(x-h)^2 + (y-k)^2 = r^2$

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29

a

i $(0, 0)$

ii 5 units

iii 10 units

iv $x^2 + y^2 = 25$

b

i $(1, -2)$

ii 4 units

iii 8 units

iv $(x-1)^2 + (y+2)^2 = 16$

30

a $(2, -3)$ **b** $r = 5\sqrt{3}$ units **c** $r = 10\sqrt{3}$ units

31

a $x^2 + (y-6)^2 = 9$

b $\left(x - \frac{17}{2}\right)^2 + y^2 = 15$

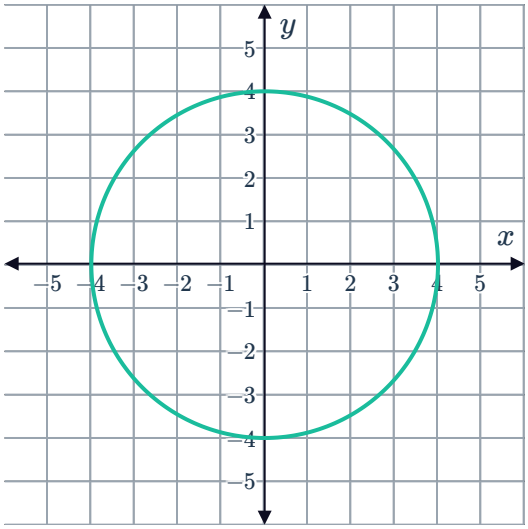
32

a

i $(0, 0)$

ii 4 units

iii



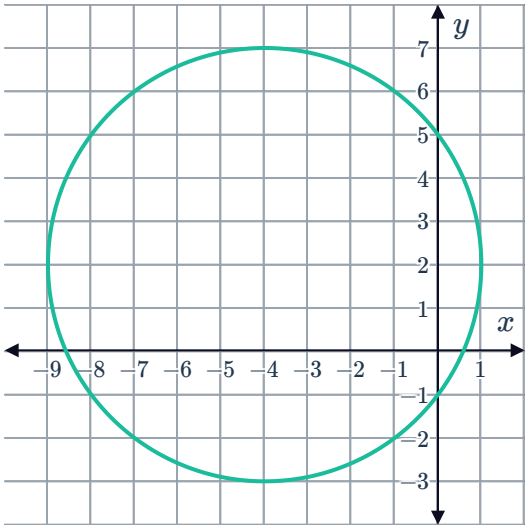
iv Domain $= [-4, 4]$
Range $= [-4, 4]$

b

i $(-4, 2)$

ii 5 units

iii



iv Domain $= [-9, 1]$
Range $= [-3, 7]$

33

a 13 units

b $(x+2)^2 + (y-5)^2 = 169$

34

a Inside **b** On **c** Outside **d** On

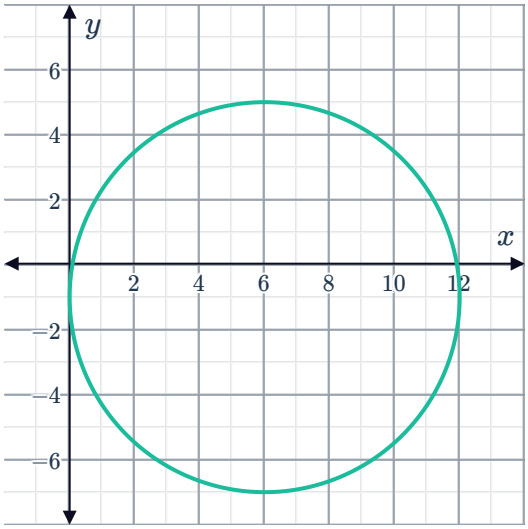
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35

a

i $(x-6)^2+(y+1)^2=36$

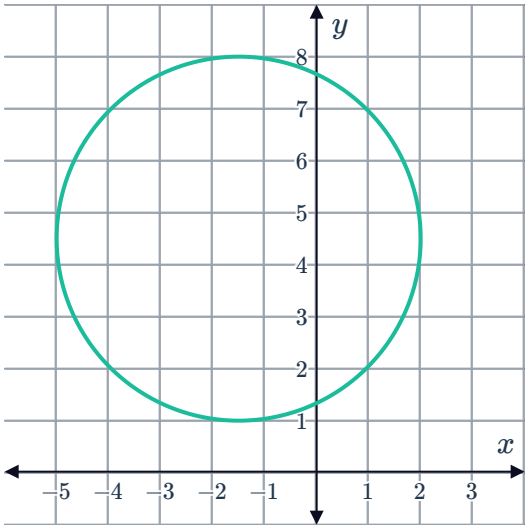
ii



b

i $\left(x+\frac{3}{2}\right)^2+\left(y-\frac{9}{2}\right)^2=\frac{49}{4}$

ii



36 $(x-25)^2+(y-30)^2=400$

37

a $(49.5, 37.5)$

b $9.15\ \text{m}$

c $(x-49.5)^2+(y-37.5)^2=83.7225$

d Domain $=[40.35, 58.65]$
Range $=[28.35, 46.65]$

38

a Domain: $[5, 19]$, Range: $[1, 15]$

b Domain: $[14, 26]$, Range: $[2, 14]$

c $5\ \text{cm}$

d $85.19\ \text{cm}^2$

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39

a $(x-4)^2+(y+3)^2=41$

b Given:

- Equation of circle: $(x-4)^2+(y+3)^2=41$
- Point: $P(8,3)$

To prove: Distance between circle center and $P(8,3)$ is greater than $\sqrt{41}$

1. Find the center of the circle

$$(x-4)^2+(y+3)^2=41 \qquad \text{Given}$$

$$C=(4,-3) \qquad \text{Equation of the circle}$$

2. Find distance between center and $C(8,3)$

$$CP=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2} \qquad \text{Distance formula}$$

$$=\sqrt{(8-4)^2+(3--3)^2} \qquad \text{Substitution}$$

$$=\sqrt{(4)^2+(6)^2} \qquad \text{Simplifying}$$

$$=\sqrt{16+36} \qquad \text{Simplifying}$$

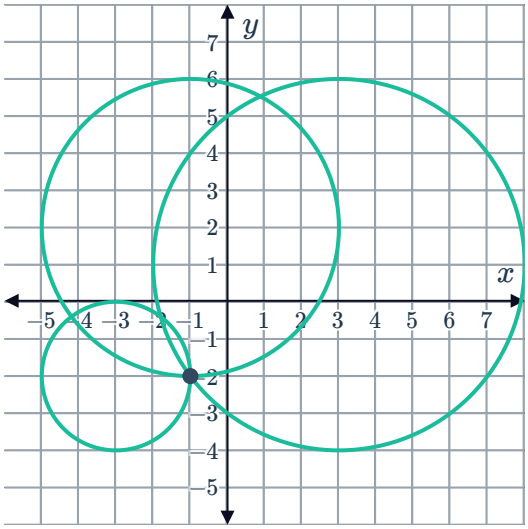
$$=\sqrt{52} \qquad \text{Simplifying}$$

3. Show distance between circle center and $P(8,3)$ is greater than $\sqrt{41}$
   $CP=\sqrt{52}>\sqrt{41}$
   Therefore, P is outside the circle.

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40

a As the point of intersection of the three circles centered at the receiving stations with radii as the distances, the epicenter is at $(-1, -2)$.



b The point $(8, -7)$ is outside the circle with center $(-1, -2)$ with radius 10, so it would not feel the earthquake.

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